Experiment No. 1

Aim : Study various applications of NLP and Formulate the Problem Statement for Mini Project based on chosen real-world NLP applications.

Subject : Natural Language Processing (NLP)

Title of Project : FinSent – Financial Sentiment Analysis for Market Trend Prediction

Selection of Topic & Justification

Topic Chosen : Real-Time Financial News Sentiment Analysis and Information Extraction for Stock Trend Prediction using Large Language Models (LLMs).

How and Why this topic was chosen

The stock market is a highly volatile environment driven primarily by information and human emotion. Every day, thousands of news articles, earnings reports, and social media posts are published. Institutional investors rely on highly expensive, proprietary software (e.g., Bloomberg Terminals) equipped with advanced Natural Language Processing (NLP) pipelines to instantly parse this textual data and trade on it. Retail investors, however, lack access to these resources.

This topic represents a pure, high-impact real-world application of NLP. By leveraging modern LLMs (like Google Gemma via OpenRouter API) and financial APIs (yfinance), we can bridge the gap between retail and institutional investors. The project challenges the NLP system to not only classify text sentiment (Positive, Negative, Neutral) but to extract the single most actionable "highlight" from complex financial jargon, making it an excellent demonstration of applied artificial intelligence.

Motivation

The primary motivation behind this project is to solve the problem of **"Information Overload"** in financial markets. Retail investors frequently make poor trading decisions based on delayed or misunderstood news. Reading through dozens of full-length financial articles to gauge market sentiment is virtually impossible for a human to do in real-time.

Furthermore, standard sentiment analysis models (like VADER or TextBlob) struggle with financial lexicon for example, the word "bullish" is highly positive in finance but neutral in standard English, while "cut" (e.g., cutting interest rates) can be positive or negative depending on context. This motivated the development of an application that uses advanced, context-aware NLP to instantly digest textual data, provide an aggregate sentiment score, and generate fundamental analyses (like SWOT) automatically.

Problem Statement

"To design and develop an NLP-powered financial dashboard that autonomously aggregates real-time market news, accurately classifies the sentiment of complex financial text, and extracts actionable insights (SWOT analysis, key highlights, and peer comparisons) to assist retail investors in making data-driven trading decisions."

Key Sub-Problems to Address :

1) **Financial Text Understanding:** Standard NLP models lack financial context. The system must accurately understand market-specific vocabulary.
2) **Contextual Highlight Extraction:** The system must read entire articles and extract the single most impactful sentence to save the user's time.
3) **Automated Fundamental Analysis:** Using generative NLP to automatically draft a Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis based on current market data and news.

Proposed Solution

The proposed solution is **FinSent**, a full-stack web application designed for retail investors.

- ✓ **Data Aggregation:** The backend (built with Python and FastAPI) will ingest real-time news headlines, stock price history, and fundamental ratios using the yfinance library.
- ✓ **NLP Pipeline:** The textual data is fed into a Large Language Model (Google Gemma-4-26b via OpenRouter). The NLP model is rigorously prompted using strict JSON schemas to perform:
  - *Sentiment Classification* (Positive/Negative/Neutral).
  - *Highlight Extraction* (Isolating the most actionable sentence).
  - *Generative Analysis* (Structuring raw data into SWOT and Peer Comparison matrices).
- ✓ **User Interface:** A highly responsive, dual-theme (Dark/Light mode) dashboard built with vanilla HTML/CSS/JS that displays live market indices, technical charts (Moving Averages), and contextually colored NLP hint boxes to provide an institutional-grade experience.

References / Citations

a) **APIs and Datasets:**

- **Yahoo Finance API (yfinance):** Sourced for real-time market data, ticker histories, and financial news headlines. *(Ran Aroussi, yfinance https://github.com/ranaroussi/yfinance)*
- **OpenRouter API:** Utilized as the gateway to access the google/gemma-4-26b-a4b-it:free Large Language Model for advanced text processing and JSON formatting.

b) Research & Literature on Financial NLP:

- Araci, D. (2019). *"FinBERT: Financial Sentiment Analysis with Pre-trained Language Models."* arXiv preprint arXiv:1908.10063. (Used as foundational inspiration for the necessity of finance-specific NLP processing).
- Kearney, C., & Liu, S. (2014). *"Textual sentiment in finance: A survey of methods and models."* International Review of Financial Analysis, 33, 171-185. (Sourced for understanding the correlation between textual news sentiment and market volatility).

c) Software Architecture:

- FastAPI Framework Documentation: https://fastapi.tiangolo.com/ (For RESTful API formulation).
- Chart.js Documentation: https://www.chartjs.org/ (For data visualization rendering).

Conclusion

The formulation of this problem statement establishes a clear, real-world application of Natural Language Processing within the financial technology sector. By selecting **FinSent**, we have identified a critical pain point for retail investors: the inability to rapidly process and interpret the overwhelming volume of financial news and data. Through the proposed integration of Large Language Models and real-time market APIs, this project aims to demonstrate how NLP can be effectively utilized to extract meaning, determine sentiment, and generate structured analytical summaries from unstructured text. This experiment successfully outlines the motivation, exact scope, and technical direction required to build an institutional-grade sentiment analysis dashboard, laying a solid foundation for the subsequent development and implementation phases of the mini-project.